

Coupled Forward ECG on Real-Geometry FEM: ASM vs. sASM Domain-Decomposition Preconditioning

Branch `cardiac-sim-fakegeo` · algorithm study (numerical linear algebra / domain decomposition)

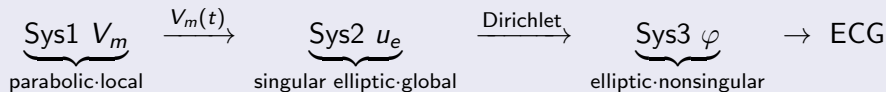
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MFEM 4.9 + PETSc 3.19 · all numbers measured on `heart.msh`

Overview: nature of the study and the main thread

Nature: a Krylov-preconditioning / domain-decomposition (Additive Schwarz) *algorithm* study. The vehicle is a coupled-PDE forward problem, but the *object of study* is how the linear systems converge faster and scale.

Pipeline



This deck focuses on: the problem definition, the mesh, and the **ASM(BASIC)** vs. **sASM(scaled/restricted)** iteration counts and **wall-clock time** on the three systems.

Problem: algebraic characterization of the three systems

Each time step solves three SPD (semi-)definite systems (`heart.msh`, Sys2 has 10085 unknowns):

System	Equation	Algebraic type	baseline iters
Sys1 V_m	$(\frac{1}{\Delta t}M + \frac{1}{2}K) V_m = b$	parabolic / mass-dominated / nonsingular	CG+ICC ~ 13
Sys2 u_e	$K_{\sigma_i + \sigma_e} u_e = -K_{\sigma_i} V_m$	elliptic / pure-Neumann singular / global	$\sim 82-96$
Sys3 φ	$K_{\sigma_o} \varphi = 0$, interface Dirichlet	elliptic / nonsingular / global	$\sim 62-66$

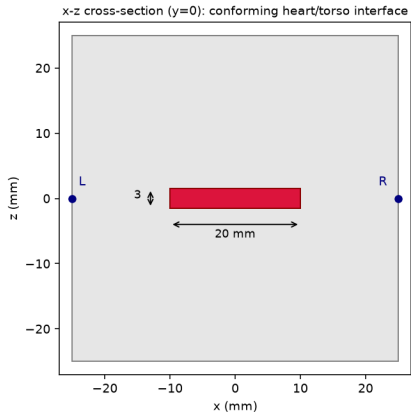
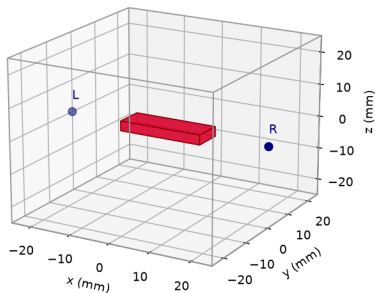
Sys2 is the only hard one (singular + elliptic + global):

- $K\mathbf{1} = 0$: null space = constants = one slow global mode $\Rightarrow$ de-mean the RHS + `MatSetNullSpace` (project the constant mode out every CG iteration).
- global elliptic $\Rightarrow$ a one-level domain decomposition propagates only one subdomain hop per iteration; more subdomains = slower (the core theme below).

Geometry: conforming heart-in-torso domains

Geometry: conforming heart-in-torso (fibers $\parallel x$; electrodes L,R at $x=\pm 25$)

Torso 50^3 mm (gray) + heart slab $20 \times 7 \times 3$ mm (red)

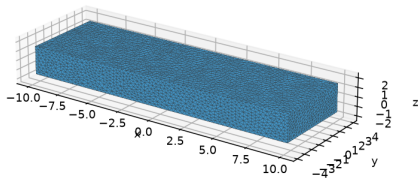


Heart slab $20 \times 7 \times 3$ mm centered in a 50^3 mm torso; **conforming** interface (Gmsh BooleanFragments $\Rightarrow$ shared interface surface). Anisotropic σ , fibers $\parallel x$; electrodes L,R at $x = \pm 25$; subdomains = MPI ranks (METIS).

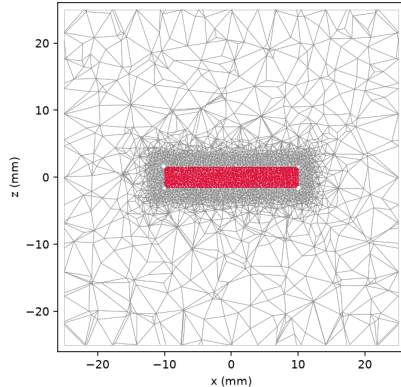
Mesh: unstructured P1 tetrahedra (real connectivity)

Unstructured P1 tetrahedral mesh (real connectivity from heart.msh / torso.msh)

Heart boundary surface mesh (9510 tris; blue=interface)



y=0 slice of the tetrahedra: torso coarse (gray) + heart fine (red)



heart	10 085 nodes	47 582 tets	9 510 tris
torso	36 229 nodes	206 647 tets	12 112 tris

interface shared dofs: 4 757; y=0 slice: heart ~ 0.5 mm, torso ~ 3 mm

Method: ASM(BASIC) vs. sASM(scaled / restricted)

One-level additive Schwarz, blocks = METIS subdomains, sub-solve = ICC(0):

$$M_{\text{ASM}}^{-1} = \sum_i R_i^{\top} A_i^{-1} R_i, \quad A_i = R_i A R_i^{\top} \text{ (overlap } O \text{ layers)}$$

The key difference — overlap *multiplicity* (over-count)

- **ASM(BASIC)**: overlap dofs get *one contribution from each* overlapping subdomain $\Rightarrow$ **over-counting** in the overlap, which corrupts the operator spectrum and can even *increase iterations with overlap*.
- **sASM**: a diagonal partition of unity $D^{-1/2}(\cdot)D^{-1/2}$ cancels the over-count ($D =$ multiplicity) $\Rightarrow$ overlap works *as intended*, iterations drop with overlap.

$$M_{\text{sASM}}^{-1} = D^{-1/2} \left(\sum_i R_i^{\top} A_i^{-1} R_i \right) D^{-1/2}$$

At $O=0$ (no overlap) there is no over-count, so **ASM** $\equiv$ **sASM**.

Iterations: ASM vs. sASM ($n_p=4$, $\text{rtol } 10^{-8}$, ICC(0))

System	O	ASM(BASIC)	sASM	note
Sys1 V_m	0	11	11	$O=0$: equal
	1	13 $\uparrow$	8 $\downarrow$	
	2	13 $\uparrow$	8 $\downarrow$	
Sys2 u_e (singular)	0	82	82	
	1	92 $\uparrow$	74 $\downarrow$	
	2	93 $\uparrow$	74 $\downarrow$	
Sys3 φ	0	62	62	
	1	88 $\uparrow$	53 $\downarrow$	
	2	94 $\uparrow$	53 $\downarrow$	

$\uparrow$ ASM iterations *rise with overlap* (over-count); $\downarrow$ sASM *falls* with overlap (overlap works correctly).

Iterations: ASM vs. sASM ($n_p=8$, more subdomains $\Rightarrow$ stronger anomaly)

System	O	ASM(BASIC)	sASM	note
Sys1 V_m	0	13	13	
	1	18 $\uparrow$	8 $\downarrow$	
	2	19 $\uparrow$	8 $\downarrow$	
Sys2 u_e (singular)	0	79	79	
	1	117 $\uparrow$	73 $\downarrow$	
	2	123 $\uparrow$	71 $\downarrow$	
Sys3 φ	0	66	66	
	1	93 $\uparrow$	55 $\downarrow$	
	2	109 $\uparrow$	53 $\downarrow$	

Most extreme: Sys3 ASM $O:0 \rightarrow 2$ **66 $\rightarrow$ 93 $\rightarrow$ 109 (+65%)**; sASM **66 $\rightarrow$ 55 $\rightarrow$ 53 (−20%)**.

Wall-clock (I): total time = setup + solve (ms, avg of 20)

$n_p = 4$

System	O	ASM	sASM
Sys1	0	3.01	3.01
	1	3.97	3.46 ↓
	2	5.10	4.43 ↓
Sys2	0	12.45	12.57
	1	16.87	13.00 ↓
	2	17.67	14.54 ↓
Sys3	0	32.08	34.59
	1	53.25	35.78 ↓
	2	63.82	41.19 ↓

$n_p = 8$

System	O	ASM	sASM
Sys1	0	5.01	4.54 ↓
	1	8.39	6.07 ↓
	2	9.81	7.37 ↓
Sys2	0	17.09	15.63 ↓
	1	31.48	23.00 ↓
	2	38.87	27.66 ↓
Sys3	0	39.97	42.12
	1	68.63	45.94 ↓
	2	110.14	54.58 ↓

Total = KSP + PCASM setup (subdomains, ICC factors) + solve. sASM overlap ($O \geq 1$) is faster; ASM overlap costs both setup and solve.

Wall-clock (II): *solve only* — setup excluded (ms, avg of 20)

$n_p = 4$

System	O	ASM	sASM
Sys1	0	1.37	1.64
	1	1.90	1.17 ↓
	2	2.03	1.31 ↓
Sys2	0	14.83	11.57 ↓
	1	12.64	10.57 ↓
	2	14.14	11.97 ↓
Sys3	0	28.59	28.34 ↓
	1	45.17	29.12 ↓
	2	56.46	31.77 ↓

$n_p = 8$

System	O	ASM	sASM
Sys1	0	2.73	3.17
	1	3.85	1.92 ↓
	2	4.87	2.98 ↓
Sys2	0	14.85	14.67 ↓
	1	29.34	23.77 ↓
	2	33.75	20.02 ↓
Sys3	0	35.00	38.65
	1	59.74	35.77 ↓
	2	82.96	40.30 ↓

Solve only = KSPSolve after KSPSetUp (setup excluded). Now the sASM/ASM time ratio tracks the *iteration* ratio: $n_p=8$ Sys3 $O=2$: ASM 82.96 vs. sASM 40.30 ms = $2.06\times = 109/53$.

Key finding: the overlap anomaly

Observation

Adding overlap to one-level **ASM(BASIC)** *raises* iterations and slows wall-clock; adding overlap to **sASM** *lowers both*.

Mechanism: overlap anomaly = over-count $\times$ inexactness

- overlap dofs counted once per subdomain $\Rightarrow$ BASIC **amplifies** in the overlap ($\|M^{-1}\|$ too large), degrading the spectrum.
- ICC(0) sub-solves are *inexact*, so the amplified error is not absorbed by an exact local solve $\Rightarrow$ iterations climb.
- the $D^{-1/2}$ partition of unity in sASM cancels the over-count exactly $\Rightarrow$ restores the correct behavior “more overlap = faster information transport”.

Conclusion: on real-geometry FEM, **scaled/restricted ASM is the correct way to use overlapping DD**; BASIC overlap is harmful here. More subdomains ($n_p:4 \rightarrow 8$) $\Rightarrow$ stronger anomaly, underscoring that *scalability* needs the correct weighting (and, later, a coarse space).

sASM parameter sweep: ICC level $L \times$ overlap O ($n_p=8$)

Cells = CG iterations (rtol 10^{-8}); **green** = optimum by *solve-only* time; **yellow** = fewest iterations.

Sys2 u_e (singular)

$L \backslash O$	$O=0$	$O=1$	$O=2$
$L=0$	79	73	71
$L=1$	62	47	46
$L=2$	55	38	35

opt: $L=1, O=1$ — 47 it, 13.5 ms

Sys3 φ (torso)

$L \backslash O$	$O=0$	$O=1$	$O=2$
$L=0$	66	55	53
$L=1$	50	36	37
$L=2$	46	30	30

opt: $L=1, O=1$ — 36 it, 39.3 ms

Optimum (measured $n_p=4$ and 8): $L=1, O=1$ for the elliptic systems

- **Overlap:** $O=1$ is best. $O=0$ is worst (no transport); $O=1$ gives the big drop; $O=2$ shaves only a few more iterations but costs more setup+apply $\Rightarrow$ solve-time usually rises $O=1 \rightarrow 2$.
- **ICC level:** $L=1$ is the sweet spot. $L=0 \rightarrow 1$ roughly halves iterations cheaply; $L=2$ gives the fewest iterations but the denser factor makes each apply+setup dearer $\Rightarrow$ time rarely improves. (Sys1, mass-dominated: $L=0, O=1$.)

Summary and outlook

- **Problem:** coupled forward ECG on three systems; Sys2 (singular pure-Neumann elliptic) is the only hard one and the target of the preconditioning study.
- **Mesh:** conforming unstructured tet P1-FEM, heart 10085 dof / 47582 tets, torso 36229 / 206647.
- **ASM vs. sASM** (measured, $n_p=4, 8$):
 - ASM(BASIC) iterations **rise with overlap** (over-count), wall-clock slower;
 - sASM iterations **fall with overlap**, and wall-clock is faster for $O \geq 1$;
 - at $O=0$ the two coincide.
- **Next steps** (implemented / planned on the same branch): strong fine level **SORAS** ($84 \rightarrow 20$, $4.2\times$), shared **Nicolaides coarse space** (scalability), cross-time **Fischer** (-12%); at scale (~ 3000 cores) a **GenEO spectral coarse space** + a telescoped scalable coarse solve.

All numbers measured in-container (MFEM 4.9 + PETSc 3.19); ECG bit-identical (verified).